CHEMISTRY

SECTION - A

Multiple Choice Questions: This section contains 20 multiple choice questions. Each question has 4 choices (1), (2), (3) and (4), out of which **ONLY ONE** is correct.

Choose the correct answer :

1. For XeO_2F_2, select the correct statement(s).

   (A) It shows see-saw shape.

   (B) Number of lone pair(s) of e^- on Xe is 1.

   (C) $\angle FXeF = 180°$ (approx.)

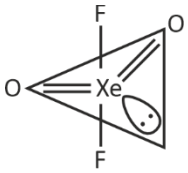
   (D) It has tetrahedral shape.

   (1) (A), (C), (D) Only (2) (A), (B) only

   (3) (A), (B), (C) only (4) (B), (C), (D) only

Answer (3)

Sol.



2. Statement I : Size of O^{2-} is smaller than F^-.

   Statement II : Electronegativity of F is more than that of oxygen.

   In the light of above statements, choose the correct option.

   (1) Both Statement I and Statement II are correct

   (2) Both Statement I and Statement II are incorrect

   (3) Statement I is correct but Statement II is incorrect

   (4) Statement I is incorrect but Statement II is correct

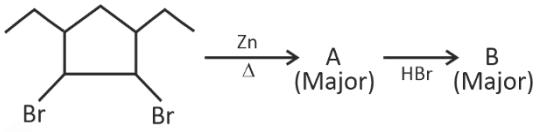
Answer (4)

Sol. Number of proton in O^{2-} is 8 and number of proton in F^- is 9 (both have $10e^-$). So, O^{2-} is larger than F^-

So, Statement I is false.

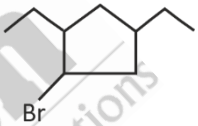
Statement II is true, F is smaller than O, so F is more electronegative than O.

3. Consider the reaction,



Choose the correct option,

(1) A is 

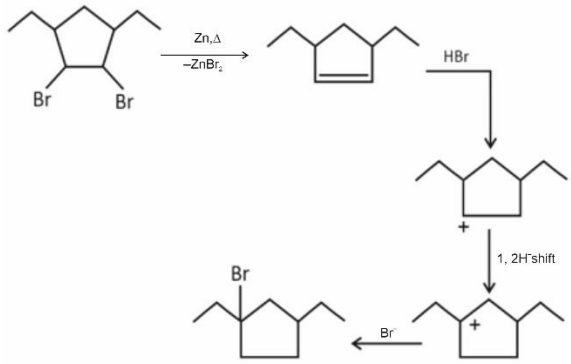
(2) B is 

(3) B is 

(4) A is 

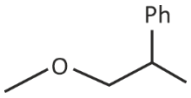
Answer (3)

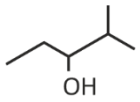
Sol.





4. Which of the following molecule gives iodoform test.

(1) 

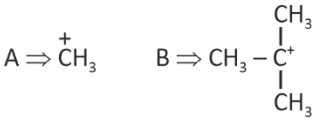
(2) 

(3) CH_3CH_2OH (4) CH_3COOH

Answer (3)

Sol. $CH_3CH_2OH \xrightarrow[NaOH]{I_2} HCOO^- + CHI_3 \downarrow$

5. Consider the following intermediates.



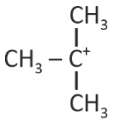
(1) B is more stable than A as it has 9 α hydrogen

(2) A is more stable than B as it has 3 α hydrogen

(3) B is more stable than A due to resonance

(4) A is more stable due to inductive effect

Answer (1)

Sol. 

$\Rightarrow$ more stable as it has 9 α-H

$\Rightarrow$ +I effect is more than B than A

6. DNA is optically active due to the presence of

(1) Purine nitrogenous base

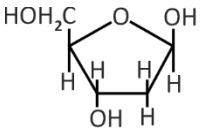
(2) Phosphate molecule

(3) D-pentose sugar

(4) L-pentose sugar

Answer (3)

Sol. DNA is optically active due to D-pentose sugar.



β-D-2-deoxyribose

7. What is the oxidation state of chromium in the product when $K_2Cr_2O_7$ reacts with acidified KI

(1) +6 (2) +3

(3) +4 (4) +5

Answer (2)

Sol. $K_2Cr_2O_7 + 6KI + 7H_2SO_4 \rightarrow Cr_2(SO_4)_3 + 4K_2SO_4 + 3I_2 + 7H_2O$

Cr in product have +3 oxidation state.

8. 250 cc of $x \times 10^{-3}$ M acidified $K_2Cr_2O_7$ solution titrates 750 cc of 0.6 M Mohr's salt completely. Value of x is

(1) 200 (2) 600

(3) 400 (4) 300

Answer (4)

Sol. meq of $K_2Cr_2O_7$ = meq of Mohr's salt

$250 \times x \times 10^{-3} \times 6 = 750 \times 0.6 \times 1$

$\Rightarrow$ x = 300

9. Two metals with work function in ratio 1 : 2, are exposed with photons of energy 6 eV. If $KE_A : KE_B$ is 2.642 : 1, then value of ϕ_A and ϕ_B (in eV) are

(1) 2.3, 4.6 (2) 1.4, 2.8

(3) 2.3, 3.6 (4) 3.2, 6.4

Answer (1)

Sol. Let ϕ_A = Y, then ϕ_B will be 2Y

Let KE_B = x, then KE_A = 2.642x

$E_i(eV) = \phi(eV) + KE(eV)$

For metal A $\Rightarrow$ 6 = Y + 2.642x ...(I)

For metal B $\Rightarrow$ 6 = 2Y + x ...(II)

From Eq (I) and (II) Y = 1.642x

Now put Y in Eq. (II)

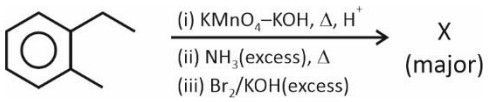
6 = 2(1.642x) + x

on solving x = 1.4, Y = 1.642 × 1.4 = 2.3

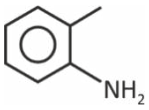
So $\phi_A = Y = 2.3\,eV, \phi_B = 2Y = 4.6\,eV$

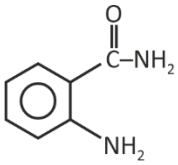


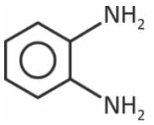
10. Consider the reaction sequence :

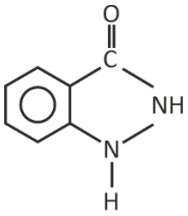


X is,

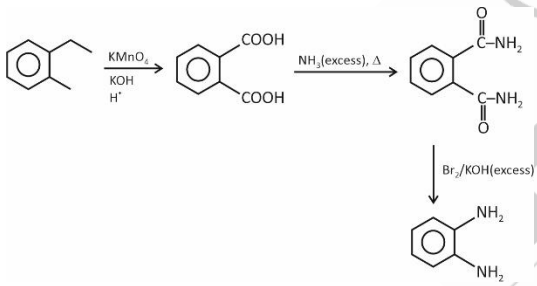
(1) 

(2) 

(3) 

(4) 

Answer (3)

Sol. 

11. Consider the following statement about complexes and its hybridisation.

A. $[CoF_6]^{3-}$; outer orbital complex, sp^3d^2

B. $[Ni(CN)_4]^{2-}$; inner orbital complex; dsp^2

C. $[Co(NH_3)_6]^{3+}$; inner orbital complex; d^2sp^3

D. $[FeF_6]^{3-}$; outer orbital complex; sp^3d^2

Choose the correct statement.

(1) A, B, C and D (2) A, B and C only

(3) A and B only (4) B and C only

Answer (1)

Sol. $[CoF_6]^{3-} \Rightarrow Co^{3+} \Rightarrow 3d^6 \Rightarrow t_{2g}^4 e_g^2$

F^- is WFL $\Rightarrow sp^3d^2$, outer orbital complex.

$[Ni(CN)_4]^{2-} \Rightarrow Ni^{2+}$ with $CN^- \Rightarrow$ SFL $\Rightarrow dsp^2$, inner orbital complex

$[Co(NH_3)_6]^{3+} \Rightarrow Co^{3+} \Rightarrow NH_3$ (SFL) $\Rightarrow d^2sp^3$, inner orbital complex

$[FeF_6]^{3-} \Rightarrow Fe^{3+} \Rightarrow F^-$ (WFL) sp^3d^2, outer orbital complex

12. Consider the following reactions

$PbO_2 + Pb \rightleftharpoons 2PbO; \Delta G_1$

$SnO_2 + Sn \rightleftharpoons 2SnO; \Delta G_2$

Select the correct option?

(1) $\Delta G_1 > 0,\ \Delta G_2 > 0$ (2) $\Delta G_1 < 0,\ \Delta G_2 > 0$

(3) $\Delta G_1 < 0,\ \Delta G_2 < 0$ (4) $\Delta G_1 > 0,\ \Delta G_2 < 0$

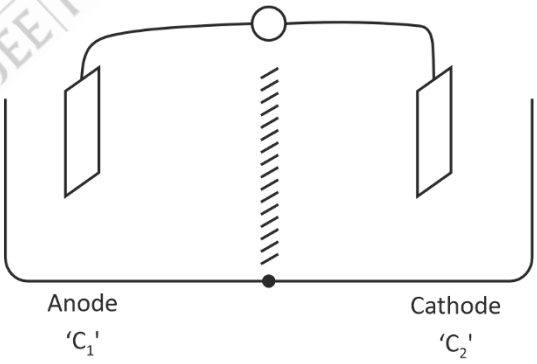
Answer (2)

Sol. Due to inert pair effect, Pb(II) is more stable than Pb(IV)

$\therefore$ PbO > PbO_2 (stability)

SnO_2 > SnO (stability)

13. Consider a galvanic cell, made up of two H_2–electrodes,



Both compartments contain the same metal electrodes. If concentrations of H^+ in anode and cathode are C_1 and C_2 respectively, then $E_{cell} > 0$ when, (p_{H_2} = 1 atm in both compartments)



(1) $C_2 < C_1$

(2) $C_2 = C_1$

(3) $C_2 > C_1$

(4) $C_2 = 0.5\ C_1$

Answer (3)

Sol. $\text{LHE}: \frac{1}{2}H_2 \rightarrow \underset{C_1}{H^+} + e^-$

$\text{RHE}: \underset{C_2}{H^+} + e^- \rightarrow \frac{1}{2}H_2$

$\text{Net}: \underset{C_2}{H^+} \rightarrow \underset{C_1}{H^+}$

$$E = E^\circ - \frac{0.0591}{1}\ln\frac{C_1}{C_2}$$

$$E_{cell} = 0.0591\ln\frac{C_2}{C_1}$$

$C_2 > C_1$ for E_{cell} = +ve.

14. A compound P with molecular formula C_6H_7N is sparingly soluble in water. However on reaction with HCl, it becomes soluble. On reaction with KOH + $CHCl_3$, it gives foul smelling compound Q. The number of different type(s) of H atoms present in P is

(1) 4

(2) 5

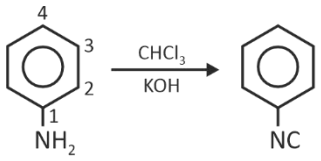
(3) 7

(4) 8

Answer (1)

Sol. C_6H_7N D.U. = 4

$C_6H_5NH_2 + HCl \rightarrow C_6H_8N^+Cl^-$ (Water soluble)



(P) (Q)

Foul smell

Number of different type of H = 4

15. X & Y are elements from group 15. The difference in electronegativity values of X and phosphorus is more than difference in electronegativity of phosphorus and Y. Element X and Y respectively are

(1) N and As

(2) As and Sb

(3) As and N

(4) As and Bi

Answer (1)

Sol. EN of N = 3.0

P = 2.1

As = 2.0

Sb = 1.9

Bi = 1.9

ΔEN(N and P) = 0.9

ΔEN (As and P) = 0.1

ΔEN (Sb and P) = 0.2

ΔEN (Bi and P) = 0.2



16. Given below are two statements.

Statement I: Potassium hypoiodite (KOI) can act as a reducing agent in reaction with H_2O_2.

Statement II: When NaOCl reacts with KI in acidic medium, NaOCl acts as oxidising agent.

In the light of above statements, choose the correct option.

(1) Both statement I and statement II are correct

(2) Both statement I and statement II are incorrect

(3) Statement I is correct but statement II is incorrect

(4) Statement I is incorrect but statement II is correct

Answer (1)

Sol. $H_2O_2 + KOI \rightarrow KI + H_2O + O_2$

$NaOCl + 2KI + 2HCl \rightarrow NaCl + 2KCl + I_2 + H_2O$

17.

18.

19.

20.

SECTION - B

Numerical Value Type Questions: This section contains 5 Numerical based questions. The answer to each question should be rounded-off to the nearest integer.

21. In estimation of chlorine by Carius method, 0.245 g organic compound gave 0.5453 g AgCl. Find percentage of chlorine in the organic compound

Answer (55)

Sol. $$\% \text{ of Cl} = \frac{\frac{0.5453}{143.5} \times 35.5}{0.245} \times 100$$

= 55.06%

22. How many of the following complex(es) have unpaired electrons

$[Ni(CO)_4]$, $[NiCl_4]^{2-}$, $[PtCl_4]^{2-}$, $[PdCl_4]^{2-}$

Answer (1)

Sol. $[Ni(CO)_4] = 3d^{10} \Rightarrow sp^3$

n = 0

$[PtCl_4]^{2-} \Rightarrow 5d^8 \Rightarrow dsp^2$

n = 0

$[NiCl_4]^{2-} \Rightarrow 3d^8 \Rightarrow sp^3$

n = 2

$[PdCl_4]^{2-} \Rightarrow 4d^8 \Rightarrow dsp^2$

n = 0

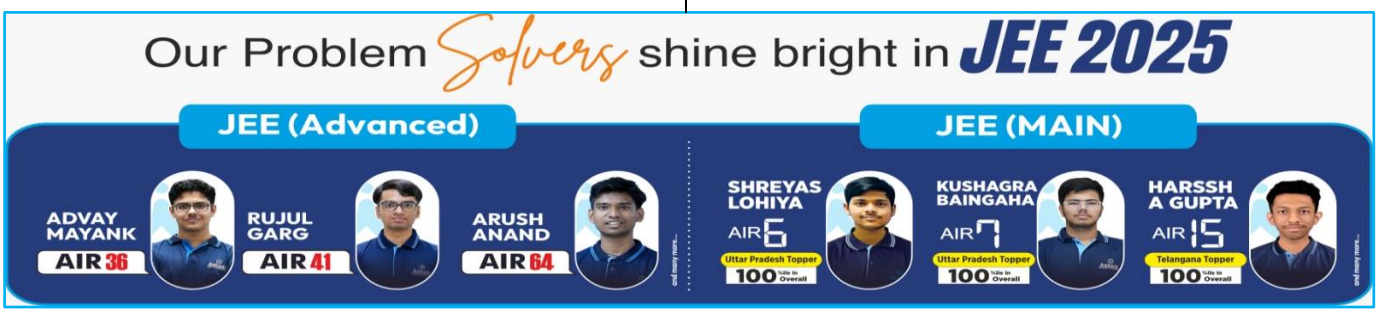
23. An ideal solution is formed by mixing 3 mole of A and 1 mole of B and the vapour pressure of solution is found to be 500 mm Hg. After further addition of 1 mole A, vapour pressure of solution becomes 520 mm Hg. Find P_A^o.

Answer (600)

Sol. $P_S = X_A P_A^o + X_B P_B^o$

$$500 = \frac{3}{4} \times P_A^o + \frac{1}{4} \times P_B^o$$

$2000 = 3P_A^o + P_B^o$ ...(i)



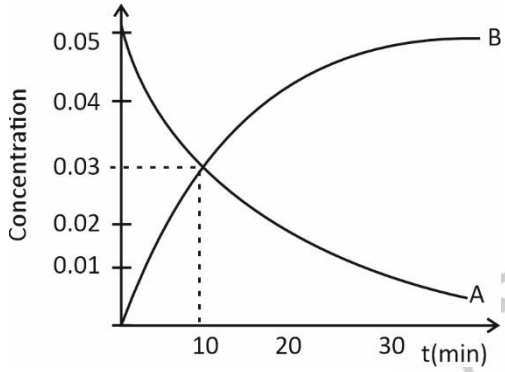
After adding 1 mole of A,

$$520 = \frac{4}{5}P_A^o + \frac{P_B^o}{5}$$

$$2600 = 4P_A^o + P_B^o \quad \ldots(ii)$$

(ii) – (i) $\Rightarrow$ 600 mm = P_A^o

24. For a reaction $\underset{(g)}{A} \longrightarrow \underset{(g)}{nB}$, a concentration vs. time curve is,



Find 10 n

Answer (15)

Sol. For the reaction,

$$-\frac{dA}{dt} = +\frac{1}{n}\frac{dB}{dt}$$

$$\frac{0.05-0.03}{10} = \frac{1}{n}\left(\frac{0.03-0}{10}\right)$$

$$\frac{0.02}{0.03} = \frac{1}{n}$$

n = 1.5

25. Consider the following reaction

$$X_2(g) + Y_2(g) \rightleftharpoons 2Z(g)$$

3 mol of X_2, 3 mol of Y_2 and 9 mol of Z are present at equilibrium and the volume of container is 1L. If 10 mol of Z is added at equilibrium, calculate the number of moles of Z at new equilibrium

Answer (15)

Sol. at eq^n $\underset{3\,mol}{X_2(g)} + \underset{3\,mol}{Y_2(g)} \rightleftharpoons \underset{9\,mol}{2Z(g)}$

$$K_{eq} = \frac{(9)^2}{3\times 3} = \frac{9\times 9}{3\times 3} = 9$$

when 10 mol of Z is added reaction moves in backward direction

$$\underset{3+x}{X_2(g)} + \underset{3+x}{Y_2(g)} \rightleftharpoons \underset{19-2x}{2Z}$$

$$9 = \frac{(19-2x)^2}{(3+x)^2}$$

$$3 = \frac{19-2x}{3+x}$$

9 + 3x = 19 – 2x

5x = 10

$$x = \frac{10}{5} = 2$$

moles of Z at new equilibrium

= 19 – 2 × 2

= 15



MATHEMATICS

SECTION - A

Multiple Choice Questions: This section contains 20 multiple choice questions. Each question has 4 choices (1), (2), (3) and (4), out of which **ONLY ONE** is correct.

Choose the correct answer :

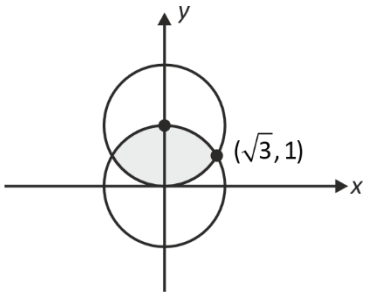
1. The area (in square units) between the curves $x^2 + y^2 = 4$ and $x^2 + (y-2)^2 = 4$ is

(1) $\frac{8\pi}{3} - 2\sqrt{3}$ (2) $\frac{8\pi}{3} + \sqrt{3}$

(3) $\frac{4\pi}{3} - 2\sqrt{3}$ (4) $\frac{4\pi}{3} + 2\sqrt{3}$

Answer (1)

Sol.



$$A = 2\int_0^{\sqrt{3}} \left(\sqrt{4-x^2} - \left(2 - \sqrt{4-x^2}\right)\right)dx$$

$$= 2\left(\left[x\sqrt{4-x^2} + 4\sin^{-1}\left(\frac{x}{2}\right)\right|_0^{\sqrt{3}} - 2(\sqrt{3})\right]\right)$$

$$= 4\left(\frac{2\pi}{3} - \frac{\sqrt{3}}{2}\right) \text{sq. units} = \frac{8\pi}{3}$$

2. Number of ways to distribute 6 identical oranges among 4 persons such that each gets at least one orange is

(1) 8

(2) 10

(3) 12

(4) 13

Answer (2)

Sol. $x_1 + x_2 + x_3 + x_4 = 6, \quad x_i \geq 1$

$\Rightarrow x_1' + x_2' + x_3' + x_4' = 2 \quad x_i' \geq 0$

$\Rightarrow$ Number of ways $= {}^{2+4-1}C_{4-1}$

$= {}^5C_3 = 10$ ways

3. The minimum value of $\cos^2\theta + 6\sin\theta\cos\theta + 3\sin^2\theta + 3$

(1) -1 (2) 1

(3) $5 + \sqrt{10}$ (4) $5 - \sqrt{10}$

Answer (4)

Sol. $\cos^2\theta + 6\sin\theta\cos\theta + 3\sin^2\theta + 3$

$4 + 3\sin 2\theta + 2\sin^2\theta$

$4 + 3\sin 2\theta + 1 - \cos 2\theta$

$E = 5 + 3\sin 2\theta - \cos 2\theta$

$E_{\min} = 5 - \sqrt{10}$

4. Let $A = \{1, 2, 3, \ldots, 9\}$; xRy iff $x - y$ is multiple of 3.

S_1 : Number of elements in R is 36

S_2 : R is equivalence relation

(1) S_1 & S_2 both are correct

(2) S_1 is correct but S_2 is not correct

(3) S_2 is correct but S_1 is not correct

(4) S_1 & S_2 both are incorrect

Answer (3)

Sol. S_1 : $(x, y) \equiv (3k_1, k_2)$ OR $(3k_1 + 1, 3k_2 + 1)$ OR $(3k_1 + 2, 3k_2 + 2)$

$\equiv 3 \times 3 + 3 \times 3 + 3 \times 3$

$= 27$ elements

S_1 is incorrect.

R is reflexive, symmetric as well as transitive relation.

as $(x, x) \in R \qquad \forall x \in A$

$(x, y) \in R \rightarrow (y, x) \in R$

$(x, y) \in R$ and $(y, z) \in R \Rightarrow (x, z) \in R$.

